What the Evidence Does and Does Not Show

— the science around The Field Trilogy, graded

JOSE GUDE, MD

A synthesis essay · physics · biology · consciousness

Contents

— On convergence — what this kind of argument can and cannot do

1. The question that is still open
2. How we got here
3. The anomalies, weighed
4. Physics I: the world is not locally real
5. Physics II: is spacetime the bottom layer?
6. Physics III: is there a smallest scale?
7. Physics IV: time-symmetric formalisms and delayed choice
8. The simulation argument
9. Consciousness-first proposals
10. The body: what is measured, and what is metaphor
11. Where the novels stand, and where the author stands
12. What is actually shifting
13. What can be claimed, and what is wagered

A SYNTHESIS · PHYSICS · BIOLOGY · CONSCIOUSNESS

What the Evidence Does and Does Not Show — the science around *The Field Trilogy*, graded

The Field Trilogy and Fragile Light are novels built on a premise: that consciousness is not produced by the brain but received through it. This page asks a different question from the novels. How much of the science the books draw on actually bears on that premise, and how strongly? Some of it is among the best-established physics there is, and genuinely strange. None of it decides whether brains produce consciousness or receive it. The anomaly literature is real and unresolved. The receiver model itself remains a speculative interpretation — a good premise for fiction, and an open question for science.

Each claim below carries a label, using the same vocabulary as the author's *Biological Individuation Theory*. Revised October 2026; an earlier version of this page made stronger claims than the evidence supports, and where it was wrong, the text below says so.

ESTABLISHED Supported by the evidence cited.

HYPOTHESIS A testable proposal, not yet confirmed.

OPEN Unresolved; the evidence is mixed or missing.

SPECULATIVE An interpretation the available evidence does not test.

NOVELS' PREMISE What the fiction assumes in order to tell its story.

SECTION

On convergence — what this kind of argument can and cannot do

METHOD

The earlier version of this page argued that *the conclusion is the convergence itself*: that no single result proves anything, but the pattern formed by all of them is harder to dismiss than any one of them. It deserves to be stated exactly, including how far it can go, because it can also mislead.

That is a real form of inference and it has a name. William Whewell called it the **consilience of inductions** in 1840: when inductions drawn from entirely different classes of fact "jump together," the agreement carries evidential weight that no single class could supply. Darwin's *Origin* is the classic example, as later philosophers of science have read it — his case rested not on any one observation but on biogeography, embryology, the fossil record, and artificial selection all pointing the same way. The reason it works is arithmetic: independent lines *multiply* rather than add. Three lines that each had a thirty per cent chance of agreeing with the thesis by coincidence give roughly three per cent for all three coinciding.

But that arithmetic depends entirely on one word, and this page would rather press on it than hope the reader doesn't.

Independence is doing all the work

Correlated errors do not cancel. They compound, and from the inside compounding looks exactly like confirmation. If several sources share a background assumption, a methodology, or simply a social world, their agreement measures the shared thing rather than the world.

Some of the convergences invoked on this site are less independent than the word suggests, and the honest move is to say which. **Bernardo Kastrup and Michael Levin converge on autopoiesis as a necessary condition — but they converge in dialogue with one another**, and it is that dialogue this site cites as the evidence. Two people reasoning together toward agreement is a weaker datum than two people arriving separately and discovering they had agreed. **Andrés Gómez Emilsson explicitly nests Friston's free-energy principle inside his own framework**, so the topological and statistical boundary criteria are not two witnesses but one account and its extension. Emilsson and Levin were in the same room at the same conference. Consciousness studies is a small field, and its participants read each other.

None of this voids the convergences. Dialogue can surface real structure, and a framework that successfully absorbs another is telling you something. But the discount is real, and a reader who noticed it unaided would be right to discount everything else on the page along with it. Better to apply it here.

Few of the convergences on this site pass the test cleanly. The quantum-gravity approaches that suggest a minimal length are theoretical frameworks built on shared starting assumptions, not independent observations. Contemplative tradi-

tions separated by geography and centuries do describe similar structures, but they share human neurology and broadly similar practices, which are common causes of similar reports. Agreement of that kind is worth explaining; it is weaker evidence than it first appears.

A curated list converges by construction

Everything in Reading is there partly because it fit. That is what a reading list is. The convergence a reader observes in a bibliography is therefore partly a fact about the bibliography, and no amount of additional agreeing sources fixes this — more agreement from a filtered pool is not more evidence.

The only corrective is to state what disconfirmation would look like and then to go and look. This site keeps the strongest opposing positions on the shelf deliberately: Daniel Dennett as "the position any field hypothesis has to answer," Anil Seth's predictive-generator account of perception as "worth knowing what the trilogy is arguing against," the retinal blind spot as evidence that perception is constructed rather than received. Those are not gestures toward balance. They are the method functioning, and they are the reason the rest is worth reading.

The distinction that matters most: criterion versus conclusion

This is the objection this page most wants to pre-empt, because it is the strongest one available against it.

Three formal vocabularies converge on a single criterion for what makes a subject: Maturana and Varela's *autopoiesis* (a self-produced boundary, maintained metabolically), Friston's *Markov blanket* (a statistical screen, maintained against dissolution), and Emilsson's *topological pocket* (a region closed by the geometry of a field). Biology, information theory, and topology, arriving at: *a maintained boundary is what individuates a subject.*

That convergence is genuine and it is impressive. It is also **compatible with straightforward physicalism**. Friston is arguably a physicalist; Emilsson calls himself one outright. Nothing in the three-way agreement requires that anything is being *received* from anywhere.

So the convergence supports **the gate** — that substrate boundaries are decisive for whether there is a subject at all — and it does not, by itself, support **the field**. Those are different claims. The novels assume both. The author's nonfiction holds only the first, and only as a hypothesis to be tested. Letting agreement about the first quietly stand in as evidence for the second would be the precise failure mode consilience is prone to, and readers should hold the page to the distinction wherever it appears.

What convergence is actually for

It shifts priors and it allocates attention. It tells you where the next decade of work is worth spending. It does not prove, and this page does not claim it does.

The line between consilience and confirmation bias is whether the convergence *terminates in a test* or terminates in satisfaction. A pattern that leaves you pleased has done rhetorical work. A pattern that generates a prediction capable of failing has done epistemic work.

The test the author's nonfiction sets itself is narrower than anything the receiver model offers, and that is its strength. *Biological Individuation Theory* asks whether independently measured organism-level organization adds reliable predictive value for the boundary, unity and continuity of experience beyond adequately specified neural measures, on held-out data and against a margin fixed in advance. If well-powered studies repeatedly find that it adds nothing, the theory weakens, and it should be allowed to. (For the novels' own version of a test, see Why biology? §4.) That commitment is what separates a research programme from a conviction, and it is why the material below is offered as a map of what is known and unknown, not as a case for a conclusion.

SECTION 1

The question that is still open

OPEN In 1995 David Chalmers, in *Facing Up to the Problem of Consciousness*, separated two kinds of problem. The *easy* problems — how the brain integrates information, discriminates stimuli, controls behaviour and reports on its own states — are difficult but tractable by the ordinary methods of neuroscience. The *hard* problem asks why any of that processing is accompanied by experience at all: why there is something it is like to see red, or to hear a suspended chord resolve. Thomas Nagel's *What Is It Like to Be a Bat?* (1974) and Frank Jackson's "Mary" thought experiment (1982) are the classic statements of the same gap.

Thirty years on, the gap has not closed, and it is fair to say so. It is equally fair to say what that does and does not license. The hard problem is a problem for *every* account of consciousness, not only for the view that brains produce it. A view on which the brain receives or filters consciousness inherits a version of it: why should any field be experiential, and how does a broad field become this particular someone? Biological Individuation Theory states that second difficulty precisely (BIT §4.13): a filter can explain how something broad becomes narrow; it does not yet explain how something non-personal becomes a person. An unsolved problem is not evidence for whichever answer one prefers.

There is a narrower question inside the hard one, and it is the one the author's nonfiction pursues: not why there is experience, but why experience is *someone's* — what fixes the bound-

ary, unity and continuity of one perspective. That question can be investigated without first deciding the metaphysics. Read the Chalmers explainer →

SECTION 2

How we got here

The view that minds depend on brains is not as old as it can seem. For much of human history the default in many cultures was closer to the reverse: Plato, the Upaniṣads, Plotinus, the Sufi metaphysicians, Vedānta and the Christian mystics, in very different ways, treated mind or spirit as more basic than matter. That history is worth knowing. It is not evidence about which view is true; long-held views have been wrong before, in both directions.

The modern settlement has seventeenth-century roots. Galileo separated the *primary* qualities physics could quantify (extension, motion, number) from the *secondary* qualities — colour, taste, sound — which he placed in the perceiver. The move made mathematical physics possible. It also set aside, rather than explained, the felt qualities of experience, and the hard problem is in part the bill for that bracket coming due.

Twentieth-century physics then made the physical world far stranger than the clockwork picture that grew out of the seventeenth century: quantum indeterminacy, entanglement, the relativity of simultaneity. Those results are real, and some of them are described below. What they did not do is show that consciousness is fundamental. Mainstream physics remains silent on consciousness, and most working physicists read their results without any reference to minds. Where the sections below discuss physics, the honest question is narrower: does a given result bear on consciousness at all, and if so, how?

SECTION 3

The anomalies, weighed

The clinical and phenomenological literature contains phenomena that do not fit comfortably inside simple accounts of how brains produce minds. They deserve study. They also deserve an accurate statement of how strong the evidence is, which varies a great deal from one item to the next. As BIT puts it (§14.1), anomalies are not the foundation, and what is unexplained is not thereby evidence for one's preferred explanation.

- OPEN **Terminal lucidity.** Nahm and colleagues (2012) collected a century of reports in which patients with severe dementia or chronic psychiatric illness briefly regain coherent speech and recognition shortly before death. A 2019 paper by a workgroup convened by the US National Institute on Aging (Mashour, Frank, Batthyány and colleagues, *Alzheimer's & Dementia*) named it *paradoxical lucidity* and called for systematic study; Batthyány's *Threshold* (2023) is the book-length treatment. The phenomenon is reported often enough by clinicians and families that it is unlikely to be pure invention. But the evidence is still mostly retrospective and anecdotal, the frequency is unknown, and no mechanism has been tested. "Not explained" means not yet explained, not inexplicable. Read the explainer →
- SPECULATIVE **Children who report previous lives.** Ian Stevenson's programme at the University of Virginia (1961–2007), continued by Jim Tucker, documented some 2,500 cases of young children describing the lives of deceased people, a subset with birthmarks said to match fatal

wounds. The archive is large and was collected with care, and its best cases are hard to explain away. The interpretation is another matter: the cases are investigated after the fact, often after families have met, and normal transmission is difficult to exclude. Paul Edwards (*Reincarnation: A Critical Examination*, 1996) is the standard critique; Stephen Braude (*Immortal Remains*, 2003) the strongest philosophical defence. Note also that even a genuine transfer of information would not by itself show that a *person* continued (BIT §14.5: information is not a person). Read the explainer →

- OPEN **Near-death experiences during cardiac arrest.** NDEs are common, structured and often life-changing; that much is well documented (van Lommel, Greyson, Parnia). The specific claim that patients perceive things they could not have perceived has been tested prospectively. The AWARE study (Parnia and colleagues, 2014) placed hidden images on shelves in resuscitation areas. Among 2,060 arrests, one patient recalled auditory details that matched events about three minutes into resuscitation; but that arrest took place where no image had been placed, so the images were never actually tested. AWARE II (2023) found that no patient identified a hidden visual image, and one of 28 identified a hidden auditory stimulus. It also recorded near-normal EEG patterns emerging as late as 35–60 minutes into CPR, a finding every account of consciousness has to take seriously. The prospective studies have so far not delivered the decisive result the hypothesis needs.
- ESTABLISHED **Severe hydrocephalus with preserved function.** Patients with greatly enlarged ventricles and thinned cortical mantle can live ordinary lives. The best-

documented case is the French civil servant described by Feuillet and colleagues in *The Lancet* (2007), with an IQ of 75 and a family and job. John Lorber's more dramatic figures (the mathematics student with "virtually no brain") reached the public through a 1980 *Science* news article and were never published as a formal case series with imaging; they should not be quoted as if they were. What these cases establish is the remarkable plasticity of a brain that develops under slowly rising pressure. They do not show that the brain is unnecessary.

- ESTABLISHED **Acquired savant syndrome.** Darold Treffert documented rare cases in which new artistic, musical or mathematical capacities follow injury or stroke. The phenomenon is real and rare. The leading explanations are neurological — release from inhibition, network reorganization — and remain hypotheses. Read the explainer →

- ESTABLISHED / SPECULATIVE **Contemplative phenomenology.** Long-term meditators across traditions describe recurring structures — awareness without thought, the thinning or loss of a felt self — and neuroimaging (Davidson, Lutz, Brewer) has tracked correlates of these states with growing precision. The reports and correlates are established. The reading of them as a narrowing filter opening onto something received is an interpretation the data do not test. Read the Teresa of Ávila explainer →

- HYPOTHESIS **Psychedelic states.** Carhart-Harris's early imaging (2012) found that psilocybin reduced activity and connectivity in default-mode hubs while subjects reported expanded experience; later work has qualified and

complicated that picture, and the "entropic brain" account remains a hypothesis. The reading that a quieter brain lets "more of the field come through" is one interpretation; that the brain's self-modelling machinery is being disrupted is another, and the data do not yet choose between them.
Read the explainer →

Taken together, these cases show that the relation between brain and experience is not a simple dial on which less brain means less mind. That is a real and interesting finding. It is not the same as showing that a damaged or quieted brain lets a signal through. Two cautions apply. First, the list is selected: the far larger body of evidence — anaesthesia, lesions, stimulation, the dependence of every specific content of experience on specific neural activity — runs the other way, and any view must account for it. Second, production models are not limited to "monotonic damage-to-function"; disinhibition and network reorganization are ordinary neuroscience. Aldous Huxley's image of the brain as a reducing valve is a striking metaphor, and BIT keeps what is useful in it while rejecting it as a theory (§4.13, §5.6). It is not a conclusion the data force.

SECTION 4

Physics I: the world is not locally real

ESTABLISHED In 1935 Einstein, Podolsky and Rosen argued that quantum mechanics must be incomplete: the correlations between separated particles should be explained by properties fixed at the source. In 1964 John Bell showed that any theory in which outcomes reflect pre-existing local properties must satisfy an inequality that quantum mechanics violates. The question became experimental. Clauser and Freedman (1972), Aspect (1982) and, by 2015, loophole-free tests by several groups found the violation. The 2022 Nobel Prize went to Aspect, Clauser and Zeilinger for this programme. Read the explainer →

That is one of the most secure results in physics: no theory that is both local and realist in Bell's sense reproduces the observations. What follows from it is still argued. Bohmian mechanics keeps definite values and accepts non-local influences; Copenhagen-style and QBist readings give up pre-existing values; many-worlds keeps locality by giving up single outcomes; superdeterministic models deny the statistical independence of measurement settings. All of them agree with experiment. None of these mainstream options requires consciousness; the minority readings that do invoke it (von Neumann and Wigner, Stapp) are not singled out by any experiment. And entanglement cannot be used to send a signal.

So Bell tells us the world is stranger than classical intuition. It does not tell us that mind is fundamental, and it should not be cited as if it did.

SECTION 5

Physics II: is spacetime the bottom layer?

ESTABLISHED (as theory) In the 1970s Bekenstein and Hawking showed that a black hole's entropy is proportional to the area of its horizon, not its volume. The result is accepted throughout theoretical physics, although Hawking radiation itself has never been observed.

HYPOTHESIS From that scaling, 't Hooft and Susskind proposed the *holographic principle*: the physics of a region may be fully described by information on its boundary. Maldacena's AdS/CFT correspondence (1997) gave it a precise form — a gravitational theory in a particular curved spacetime (anti-de Sitter) equivalent to a quantum field theory on its boundary. AdS/CFT is one of the most productive ideas in modern physics. Our universe, however, is not anti-de Sitter, and whether holography applies to it is an open research question.

SPECULATIVE Maldacena and Susskind's ER=EPR conjecture (2013) proposes that entangled systems are connected by microscopic, non-traversable wormholes. It is a serious conjecture by serious physicists and it is not established. The broader programme of deriving spacetime from entanglement is active and promising, and also unfinished. Wheeler's "It from Bit" is a philosophical slogan, not a result. Donald Hoffman's claim that spacetime is a perceptual interface used by networks of conscious agents goes much further than any of this physics and should be read as a philosophical proposal. Read the Hoffman explainer →

The fair summary: several research programmes suggest that spacetime may not be fundamental. None of them has been confirmed, and none of them, if confirmed, would by itself say anything about consciousness.

SECTION 6

Physics III: is there a smallest scale?

HYPOTHESIS The Planck length (about 1.6×10^{-35} m) is where quantum mechanics and gravity are both expected to matter. A well-known thought experiment suggests that probing distances below it would require so much energy that the probe would form a black hole, which is why many physicists expect a minimal measurable length. Sabine Hossenfelder's 2013 review in *Living Reviews in Relativity* surveys how a minimal length appears in several approaches to quantum gravity — generalized uncertainty principles, string theory, loop quantum gravity, and others — and also how the idea is formulated differently, and sometimes evaded, in each.

HYPOTHESIS In loop quantum gravity, Rovelli and Smolin (1995) showed that area and volume come in discrete quanta. That is a result *within* loop quantum gravity, which has not been tested. ESTABLISHED (as theory) Bekenstein's bound on the information a finite region can hold is widely accepted.

OPEN There is no experimental evidence that spacetime is grainy. Observations of distant gamma-ray bursts have constrained some ways in which a Planck-scale structure could have shown up (for example, Fermi's GRB 090510 measurement, 2009), and found nothing. The image of a universe "rendered" at finite resolution, with the Planck length as its pixel size, is a metaphor. It is a helpful one for fiction; it is not how the physics literature describes the question. Read the Planck-scale explainer →

SECTION 7

Physics IV: time-symmetric formalisms and delayed choice

ESTABLISHED (as formalism) Aharonov and Vaidman's two-state vector formalism describes a quantum system between two measurements using one state evolving forward from the first and one evolving backward from the second. It makes the same predictions as standard quantum mechanics. OPEN Whether its backward-evolving state is physically real, or a bookkeeping device, is a question of interpretation. Read the explainer →

ESTABLISHED Wheeler's delayed-choice experiment, proposed in 1978, has been carried out with photons (Jacques and colleagues, 2007) and single atoms (Manning and colleagues, 2015). The results agree exactly with quantum mechanics. They are often summarized as "a later choice changes what the particle did earlier." That summary is one interpretation, and most physicists reject it: standard quantum mechanics predicts the results without any influence running backward in time, because it never assigns the particle a definite "wave" or "particle" history in the first place. SPECULATIVE Retrocausal interpretations exist and are respectable; the experiments do not single them out. Read the explainer →

HYPOTHESIS (partly proven, within one theory) Arkani-Hamed and Trnka's amplituhedron (2014) computes particle-scattering amplitudes in a particular theory — planar N=4 super-Yang–Mills, a highly symmetric model, not the physics of our world — as the volume of a geometric object, from which locality and unitarity emerge rather than being assumed. Parts of the correspondence have been proven; the full picture is

still conjectural even within that theory. OPEN Whether something like it holds for the real world is unknown. Read the explainer →

Together these show that the textbook picture of time and locality may not be the deepest description. They do not show that the future constitutes the past, and the earlier version of this page was wrong to say so.

SECTION 8

The simulation argument

SPECULATIVE Nick Bostrom's 2003 argument is a trilemma: either civilizations rarely reach the ability to run detailed simulations of minds, or they rarely choose to, or we are probably in one. It is a valid argument about probabilities given its premises, and it has earned a place in academic philosophy. Its premises — in particular, that simulated minds would be conscious — are exactly what is in dispute.

Writers such as Rizwan Virk have argued that our universe "looks like" a simulation: a minimum length as voxel size, measurement as rendering on demand, conservation laws as compression. Each of these is an analogy, not a finding. The Planck length is not known to be a lattice (and a simple lattice would conflict with tested features of relativity); quantum measurement has many interpretations, most with no rendering in them. Physicist James Gates and colleagues found that certain diagrams used to study supersymmetry are related to error-correcting codes; that is a genuine mathematical observation about a theory that has not been experimentally confirmed, and it says nothing about who or what runs the universe. Read the companion essay →

The simulation hypothesis is a fine engine for fiction and a legitimate philosophical question. It is not supported by physics in any way that would distinguish it from its alternatives.

SECTION 9

Consciousness-first proposals

A number of researchers have proposed that consciousness, or something like it, is fundamental. They are worth reading closely, and the novels draw on several of them. With one partial exception (Orch-OR, below), they are also proposals that have not been tested in a way that could separate them from their rivals.

- SPECULATIVE **D'Ariano and Faggin** propose that consciousness and free will are features of quantum information itself, not products of matter. The framework is carefully argued. It does not yet make predictions that a conventional physics would fail. Explainer →
- SPECULATIVE **Maria Strømme** (2025, *AIP Advances*) offered a mathematical formalization of consciousness as a fundamental field, with individual minds as local excitations. The paper was retracted by the journal in May 2026 over concerns about its scientific validity, including that a central element of the theory had no associated measurable quantity and its predictions could not be tested. The earlier version of this page called it a peer-reviewed anchor for the field framing; that is no longer true, and the retraction is itself a lesson in why formalizing an idea is not the same as testing it. Explainer →
- SPECULATIVE **Donald Hoffman's** evolutionary argument that perception tracks fitness rather than truth is a serious result in its own terms; his further claim that reality is a network of conscious agents is a metaphysical proposal. Explainer →
- HYPOTHESIS **Penrose and Hameroff's Orch-OR** holds that consciousness depends on quantum processes in neuronal

microtubules ending in gravity-related collapse. It is unusual among these proposals in being testable, and parts have been tested. Laboratory work has reported microtubule resonances and, more recently, collective optical effects in microtubule networks, and an animal study found that a microtubule-stabilizing drug delayed anaesthetic-induced unconsciousness. None of this demonstrates quantum coherence functioning in a living brain, and an underground experiment at Gran Sasso (Donadi and colleagues, 2021) ruled out the simplest version of the Diósi–Penrose collapse model the theory relies on. Orch-OR is a live, contested hypothesis. Explainer →

- SPECULATIVE **Bohm, Pribram and Stapp** each proposed ways mind and physics might be more intimately related than the standard picture allows — the implicate order, holographic memory, and the observer's role in quantum measurement. They remain minority interpretations. Bohm → · Pribram → · Stapp →
- SPECULATIVE **Michael Levin's "Platonic space"** proposes that bodies and minds are interfaces through which pre-existing patterns "ingress" into the physical world. It is a bold extension of a strong experimental programme (see §10), and should be kept separate from it: the experiments do not test the Platonic claim. Explainer →
- ESTABLISHED / SPECULATIVE **Contemplative convergence.** Kashmir Shaivism, Christian apophatic mysticism, Sufism, Gnostic and Kabbalistic traditions describe strikingly similar structures of contemplative experience. The similarity of the reports is real and worth explaining. Whether it reflects contact with a common ground, or a

common human neurology trained by similar practices, is not something the reports themselves can settle; scholars of mysticism have argued both sides for a century. Kashmir Shaivism → · Eckhart, the Cloud, and the Kabbalah →

Every consciousness-first view also owes an answer to a problem it tends to leave aside: how one fundamental consciousness becomes many separate subjects, each with its own boundary, history and point of view. Philosophers call this the decombination problem. BIT argues that biological organization does the bounding, and that, unlike field-level accounts such as Bernardo Kastrup's, this proposal makes predictions that can fail (BIT §7.9, §5.19): *the individual is not hidden in the ground.*

SECTION 10

The body: what is measured, and what is metaphor

The novels picture the body as a finely tuned instrument — an antenna. As an image, it carries much of their beauty. As a scientific claim, each strand has to be taken separately.

- ESTABLISHED / SPECULATIVE **Golden-ratio geometry.** The golden angle (about 137.5°) in leaf and seed arrangement is real, and has a physical explanation: Douady and Couder (1992) showed that it can emerge from a simple physical rule of growth, with each new element forming where there is most room. Other popular examples are weaker. The cochlea is a spiral but not specifically a golden one, and claims that DNA's proportions follow the golden ratio depend on which measurements are chosen (on the usual textbook figures for B-DNA, the major and minor grooves are about 22 and 12 ångströms wide, a ratio nearer 1.8 than 1.618). The idea that these geometries exist in order to receive a field is not a scientific finding. Read the φ explainer →
- ESTABLISHED **Frequency and form.** Chladni's and Jenny's vibrating plates show how standing waves organize matter into patterns. That is ordinary wave physics and it is genuinely beautiful. It is not evidence that the body receives consciousness. More on Chladni's plates →
- HYPOTHESIS / OPEN **Quantum biology.** Quantum effects do contribute to some biological functions; enzyme tunnelling is well supported, and the radical-pair mechanism for the bird compass has strong support (including the magnetic sensitivity of cryptochrome 4 in

songbirds, Xu and colleagues, 2021). The once-famous long-lived quantum coherence in photosynthesis has been substantially revised: later analysis found the long-lived signals were mostly vibrational, with electronic coherence lasting only tens of femtoseconds (Duan and colleagues, 2017). Quantum effects in olfaction remain speculative. A body that uses quantum effects is not therefore a receiver of consciousness. Read the quantum-biology survey →

- ESTABLISHED / HYPOTHESIS **Bioelectricity.** Michael Levin's laboratory has shown that patterns of voltage across tissues help control anatomy: altering them can produce planaria that keep regenerating two heads in later rounds of cutting, a state held in the tissue's voltage pattern, with no change found in its tissue structure or in the patterning genes examined. Trained planaria have been reported to show some retention of training after regenerating a new head. These are strong, important findings. Levin's broader claim, that problem-solving competence is present at every biological scale, is a productive hypothesis. Reading any of this as a "receiver at the cellular scale" goes beyond what the experiments test. Read the explainer →
- ESTABLISHED **The ear as a frequency analyser.** Békésy's Nobel-winning work showed that the cochlea separates sound into its frequencies along its length. That is how hearing works. It is not evidence that the senses are tuned to a field. Oster & binaural beats →
- HYPOTHESIS **Singing and the body.** A handful of small studies (Grape, Kreutz, Fancourt, Good and Russo, among others) report that singing can lower cortisol and that group singing affects mood and, in some studies, oxytocin. The results are mixed — performance can raise cortisol —

and the effects are modest. Read the music-and-body explainer →

There is a better question here than "is the body an antenna?", and it is testable. The coordinated, self-maintaining organization of the whole organism — its boundary, its regulation of its own interior, its bioelectric and metabolic coordination — might help determine the boundary, unity and continuity of a conscious perspective, beyond what neural measures alone predict. That is the conservative form of Biological Individuation Theory, and it is labelled for what it is: HYPOTHESIS. It could fail. The companion volume *Someone Here, Now — Measured* sets out the studies that would test it, fixes in advance what result would count against it, and keeps the stronger metaphysical claim behind a firewall: evidence that the organism shapes experience is not evidence that experience exists apart from the organism. See the works in progress → · Why biology? →

SECTION 11

Where the novels stand, and where the author stands

NOVELS' PREMISE The Field Trilogy and *Fragile Light* are fiction, and they are built on a premise: that consciousness is not produced by the brain but received through it. Fiction is a good place to take a premise all the way and feel what follows. The books do that deliberately, and they should be read that way.

Anima asks the question through cases. A physician at the Boise VA has spent twenty-four years collecting patients whose experiences defy neurological explanation, and quietly builds his case that consciousness is received rather than produced: Mr. Martinez's terminal lucidity; Mary Parker's near-death experience under prolonged cerebral hypoxia; Lucía Reyes, born with a birthmark matching her dead father's fatal wound; Eddie Cortez halting a column two hundred metres short of an IED no instrument detected. The clinical voice behind the book is set out directly in two first-person essays: A Clinical Life → · Death and Dying →

Numen takes entanglement as plot. Eight years after José's death, his son Alex finds a photograph of a fractal triangle whose angles match an unresolved chord, and in Boise encounters Sable, a bio-computational intelligence who has been there the whole time, carrying a signal below the threshold of resolution.

Limen is the companion volume: the science, philosophy and contemplative testimony underneath the first two books, set out as the framework the novels dramatize.

Fragile Light, the stand-alone novel, sets the wager in human terms: if consciousness is fundamental, then freedom becomes the variable that matters most. "Freedom did not lose. Freedom was interrupted." The book is about what survives the interruption.

The author's nonfiction takes a narrower and more careful position than the novels. *Biological Individuation Theory* is neutral on whether brains produce consciousness or individuate a more fundamental phenomenality; it treats the second possibility, Strong BIT, as speculative and keeps it behind a firewall. On one question it is not neutral: because the only demonstrated carrier of a person's continuity through every gap we know of — sleep, anaesthesia, coma — is the living organism, the claim that a person survives the organism's irreversible dissolution is, in BIT's words, *presumptively false* unless an independent carrier is demonstrated. The novels imagine otherwise. That difference is intentional: the novels are the place for the premise, and the nonfiction is the place for the evidence. Read about BIT and its companion books →

SECTION 12

What is actually shifting

Thomas Kuhn's *The Structure of Scientific Revolutions* (1962) describes how anomalies accumulate until a field reorganizes around a new framework. It is tempting to claim that consciousness science is in the middle of such a revolution. The honest description is more modest. Read the Kuhn companion essay →

- **In physics,** the foundations of quantum mechanics are again a respectable research field, and quantum gravity remains unsolved. Several programmes suggest spacetime may be emergent. None has been confirmed, and none implies anything about consciousness.
- **In consciousness science,** theories have multiplied, and the field has begun testing them against each other. The first large adversarial collaboration, between integrated information theory and global neuronal workspace theory, reported in *Nature* in 2025 that key predictions of both were challenged. That is what a healthy, unsettled science looks like. The question of production versus reception has not been put to a decisive test.
- **In biology,** developmental bioelectricity and the study of how organisms maintain themselves as individuals have become serious fields. Whether they bear on consciousness is the open question BIT is designed to test.

What has changed is not that materialism has been refuted. It is that several questions once treated as closed — what measurement is, whether spacetime is fundamental, what makes a subject one — are open again, and can be worked on carefully.

SECTION 13

What can be claimed, and what is wagered

ESTABLISHED Bell-inequality violation and the failure of local realism. Black-hole entropy scaling with area (as theory). The golden angle in plant growth and its physical explanation. Bioelectric control of anatomy. Phenomena such as acquired savant syndrome and preserved function in severe hydrocephalus. Structured contemplative and near-death reports.

HYPOTHESIS Holography beyond anti-de Sitter space. A minimal length in quantum gravity. Orch-OR. The radical-pair bird compass. Conservative BIT: organism-level organization shapes the boundary, unity and continuity of experience.

OPEN Terminal lucidity's frequency and mechanism. Whether any near-death perception is veridical. What quantum measurement is. Whether spacetime is emergent.

SPECULATIVE ER=EPR. The simulation hypothesis. Consciousness as a fundamental field. Retrocausal readings of delayed choice. Strong BIT. The receiver model.

NOVELS' PREMISE The books wager that consciousness is fundamental, that the body is an instrument through which it is received, that freedom is the act of one localized mind choosing what comes next, and that time and distance are not what they seem. The wager is what makes the stories go. Nothing on this page shows it to be true, and nothing shows it to be impossible.

An earlier version of this page argued that the evidence had moved far enough that the only remaining question was which non-materialist view fits best. That went beyond the evidence, and it has been withdrawn. The position this site

now takes is the one in *Biological Individuation Theory*: hold the metaphysics open, say what each piece of evidence actually shows, state in advance what would count against your own view, and go and look. Curiosity about the unresolved is not a weakness of this approach. It is the reason for it.

To follow the threads further: the Chalmers explainer for the hard problem; Crick & Koch for the neuroscience of consciousness this site's novels argue with; Libet and Schurger for what the readiness potential does and does not show about free will; the Bell theorem and Planck scale explainers for the physics; Michael Levin's bioelectricity for the biology; terminal lucidity for the clinical anomaly; and the strongest case against everything on this site and what would change my mind for the counter-arguments. The full citation list is in Reading. For the author's theory and its test, see the works in progress. Or simply read the novels — *Anima*, *Numen*, *Limen*, *Fragile Light* — as what they are: a premise followed all the way.

COLOPHON

This pamphlet collects the synthesis essay from josegudemd.com in a single typeset volume.

Set in Libre Baskerville at 10.5 pt over 1.55, on a 6 × 9 inch trim.

Section headings and footnote markers in Inter.

Companion to *The Field Trilogy*: *Anima* · *Numen* · *Limen*, and the standalone *Fragile Light*.

© 2026 Jose Gude, MD · Boise, Idaho